

Sample Table Structure

Before starting, create and populate this table:

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    department VARCHAR(50),  
    salary DECIMAL(10,2)  
);
```

```
INSERT INTO employees (emp_id, name, department, salary)  
VALUES  
(1, 'John', 'HR', 4000),  
(2, 'Maria', 'IT', 6000),  
(3, 'David', 'Finance', 5000),  
(4, 'Anna', 'IT', 7000),  
(5, 'Steve', 'HR', 4500),  
(6, 'Sara', 'Finance', 5500);
```

Exercises with Solutions

Retrieve all columns from the employees table

Task: Show all data.

```
SELECT *  
  
FROM employees;
```

Solution Output → Entire table with all rows and columns.

Retrieve names and salaries of employees working in IT

Task: Display only name and salary where department is IT.

```
SELECT name, salary
```

FROM employees

WHERE department = 'IT';

Result:

name	salary
Maria	6000
Anna	7000

List employees earning more than 5000, sorted by salary descending

SELECT name, department, salary

FROM employees

WHERE salary > 5000

ORDER BY salary DESC;

Result:

name	department	salary
Anna	IT	7000
Maria	IT	6000
Sara	Finance	5500

Insert a new employee record

Task: Add a new employee: *emp_id=7, name='Kevin', department='Finance', salary=4800*.

INSERT INTO employees (emp_id, name, department, salary)

VALUES (7, 'Kevin', 'Finance', 4800);

Result: Kevin is added to the table.

Increase salary of all HR employees by 5%

UPDATE employees

SET salary = salary * 1.05

WHERE department = 'HR';

Effect: John and Steve get a 5% salary increase.

Delete employees earning less than 4500

DELETE FROM employees

WHERE salary < 4500;

Effect: John is removed if his salary remains 4000 after updates.

Retrieve highest salary employee

Method 1 (ORDER BY + LIMIT):

SELECT name, salary

FROM employees

ORDER BY salary DESC

LIMIT 1;

Method 2 (MAX):

SELECT name, salary

FROM employees

WHERE salary = (SELECT MAX(salary) FROM employees);

Result: Anna (7000)

Find total salary paid to each department

SELECT department, SUM(salary) AS Total_Salary

FROM employees

GROUP BY department;

Sample Output:

department	Total_Salary
HR	8925

IT	13000
Finance	10300

Display employees whose name starts with 'S'

SELECT name, department, salary

FROM employees

WHERE name LIKE 'S%';

Result:

name	department	salary
Steve	HR	4500
Sara	Finance	5500

Show department-wise highest salary

SELECT department, MAX(salary) AS Highest_Salary

FROM employees

GROUP BY department;

Sample Output:

department	Highest_Salary
HR	4700
IT	7000
Finance	5500

Practice Tips

- Try **changing conditions** (e.g., <, BETWEEN, IN) to explore different results.
- Add more employees to test how queries handle larger datasets.
- Combine **GROUP BY** and **ORDER BY** for advanced analysis.